Core Finance
Week 4

MGMP 543

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Spring 2025

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Welcome Back! Today's Topics

- Week 3 Recap
- Risk & Return
  - Single Stock Returns
  - Risk
  - Statistics
  - Returns and Risk on Multiple Stocks
  - Efficient Frontiers & Portfolio Theory
  - Limits of Diversification

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Week 4 Introductions

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Your Name

Unpopular Opinion – something you think is over-rated/under-rated:

- TV/movie? “Friends isn’t funny”
- Musician? “Kendrick Lamar is over-hyped”
- Food? “Pineapple on pizza is the best”

Week 3 Recap

Week 3: What Did We Learn?

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- Equity is ownership of a corporation
- Types of equity: Common stock and preferred stock
- Value of equity it discounted expected future value of dividends (or free cash flows)
- Capitalization rate
- Present Value of Growth Opportunities & Market-to-Book Values
- Stock Exchanges, Stock Market Index
- Long-Run Returns and Short-run Volatility
- Price-Earnings (P/E) Ratios

Quiz Question 1. GK Company just paid a dividend of $2 per share out of earnings of $4 per share. If the book value per share is $25, what is the expected growth rate in dividends (g)?

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$g = ROE \times b$

ROE = EPS / Book Value Per Share = \$4/\$25 = 16%

Plowback: b = 1 – D/EPS = 1 - \$2/\$4 = 1 – 0.5 = 0.5

$g = 16\% \times 0.5 = 8\%$

Quiz Question 2. XYZ Corporation just paid an annual dividend of $3 per share on its common stock. This dividend is expected to grow at a 10% annual rate for two years, after which it is expected to grow at a 6% annual rate forever. If the required return is 10%, what value would you place on this stock? (Hint: the dividend one year from today is $3.30.)

Quiz Question 2. XYZ Corporation just paid an annual dividend of $3 per share on its common stock. This dividend is expected to grow at a 10% annual rate for two years, after which it is expected to grow at a 6% annual rate forever. If the required return is 10%, what value would you place on this stock? (Hint: the dividend one year from today is $3.30.)

Calculate this in three steps:

1. Calculate dividends for first two years (where they grow at 10%)
2. Calculate dividend at end of year 2 (where they start growing at 6%)
3. Sum up dividends over next two years, along with growing perpetuity for year three onwards. Discount all of these to today.

$$D_1 = 3 \times (1 + 0.1) = \$3.30$$

$$D_2 = 3.30 \times (1 + 0.1) = \$3.63$$

$$D_3 = 3.63 \times (1 + 0.06) = \$3.85$$

$$P_0 = \frac{\$3.30}{(1+0.1)} + \frac{\$3.63}{(1+0.1)^2} + \frac{\$3.85}{(0.1 - 0.06) \times (1+0.1)^2} = \$85.5454$$

Timing of Dividend Payments

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- Imagine you own a share today. But then tomorrow you sell it.
- If dividends are paid a week from now, what happens?

Timing of Dividend Payments

- Imagine you own a share today. But then tomorrow you sell it.
- If dividends are paid out a week from now, what happens?

- **Declaration Date:** Company's board announces amount and timing of dividends.
- **Ex-Dividend Date / Ex-Date**: (Ex-ante date) Investor who buys stock on ex-dividend date or after doesn't receive the next dividend payment, the seller does.
- Why? You were owner of firm who produced those earnings.
- **Record Date:** One day after ex-dividend date determines who is due dividend.
- Dividends paid out at a later '**payment date**' after record date.

Single Stock Returns

Single Stock Returns

- Stock returns have two components:
  - Dividends – often stable and predictable
  - "Capital Gains" from changes in the price of the stock – often unstable & hard-to-predict
- Rate of return (R_t) from holding stock is sum of these, as a percent of the price paid for the stock

$$Return = \frac{Dividends + Ending\ Price - Beginning\ Price}{Beginning\ Price}$$

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$$R_t = \frac{D_t + P_t - P_{t-1}}{P_{t-1}}$$

Historical (Single) Stock Returns

Observe stock returns over N days/weeks/months/quarters/years

1. **Arithmetic Mean ("Average", "Rebalanced", "Equally Weighted") Return** is an estimator of the unconditional expected stock return:

2. **Geometric Mean ("Buy and Hold") Return:**

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Historical (Single) Stock Returns

Observe stock returns over N days/weeks/months/quarters/years

1. **Arithmetic Mean (“Average”, “Rebalanced”, “Equally Weighted”) Return** is an estimator of the unconditional expected stock return:

$$\bar{R}_{Arithmetic} = \frac{R_1 + R_2 + \cdots + R_N}{N}$$

2. **Geometric Mean (“Buy and Hold”) Return:**

$$(1 + \bar{R})^N = (1 + R_1) \times (1 + R_2) \times \cdots \times (1 + R_N)$$

$$\bar{R}_{Geometric} = [(1 + R_1) \times (1 + R_2) \times \cdots \times (1 + R_N)]^{1/N} - 1$$

Q1. Last year your stock had great returns, earning you 20%. But at the end of this year your returns are down 20%.

a) What is the (annual) arithmetic mean of your stock's return?

b) What is the (annual) geometric mean of your stock's return?

Year	Stock Value	Equivalent Value By End of Year 2
0	$100	$100
1		
2		

Q1. Last year your stock had great returns, earning you 20%. But at the end of this year your returns are down 20%.

a) What is the (annual) arithmetic mean of your stock's return?

$$\bar{R}_{Arithmetic} = \frac{20 - 20}{2} = 0\%$$

b) What is the (annual) geometric mean of your stock's return?

$$\bar{R}_{Geometric} = [(1 + 0.2) \times (1 - 0.2)]^{\frac{1}{2}} - 1 = -2.02\%$$

Year	Stock Value	Equivalent Value By End of Year 2
0	$\$100$	$\$100$
1	$\$100 \times (1 + 0.2) = \$120.00$	$\$100 \times (1 - 0.0202) = \$97.98$
2	$\$120 \times (1 - 0.2) = \$96.00$	$\$97.98 \times (1 - 0.0202) = \$96.00$

Q2. Calculate (a) the total return on your stock and (b) the (annual) geometric mean on your stock

You purchase GK stock for $23.19 today and it earns the following:

Year	Price	Dividend
0	$23.19	0
1	$24.90	$0.23
2	$23.18	$0.24
3	$24.86	$0.25

Q2. Calculate (a) the total return on your stock and (b) the (annual) geometric mean on your stock

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1	$24.90	$0.23
2	$23.18	$0.24
3	$24.86	$0.25

(a) Total Return: $\frac{\$0.23 + \$0.24 + \$0.25 + \$24.86 - \$23.19}{\$23.19} = 10.31\%$

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Q2. Calculate (a) the total return on your stock and (b) the (annual) geometric mean on your stock

You purchase GK stock for \$23.19 today and it earns the following:

Year	Price	Dividend	Yearly Return
0	\$23.19	0	
1	\$24.90	\$0.23	$\frac{\$0.23 + \$24.90 - \$23.19}{\$23.19} = 8.37\%$
2	\$23.18	\$0.24	$\frac{\$0.24 + \$23.18 - \$24.90}{\$24.90} = -5.94\%$
3	\$24.86	\$0.25	$\frac{\$0.25 + \$24.86 - \$23.18}{\$23.18} = 8.33\%$

(b) (Annual) Geometric Return:

$$\bar{R}_{Geometric} = [(1 + 0.0837) \times (1 - 0.0594) \times (1 + 0.0833)]^{\frac{1}{3}} - 1 = 3.36\%$$

(Arithmetic mean would be 8.37%-5.94%+8.33% = 3.59%)

Risk

Last week, we showed how over long time horizons stocks outperform bonds, treasuries, real estate, gold, and cash…

However, we also showed how the prices of individual stocks, and stock indexes can move a lot!

Where **r** you coming from?

So far, we've taken discount **rates** as given. Where do they come from?!

$$PV = \sum_{t=0}^{\infty} \frac{C_t}{(1+\boldsymbol{r})^t}$$

In the real world, the discount rate isn't given to you!

How to determine the correct discount rate?

1. Use the capitalization rate (last week's class)

$$r = \frac{D_1}{P} + g = \text{dividend yield + growth rate}$$

2. Use historical data (this week's class)

3. Use asset pricing models (next week's class)

Would you rather?

A: $10,000 with 100% probability

B: $0 with 50% probability, $20,000 with 50% probability.

Would you rather?

A: \$10,000 with 100% probability

E[A] = \$10,000

B: \$0 with 50% probability, \$20,000 with 50% probability.

E[B] = \$0 x 0.5 + \$20,000 x 0.5 = \$10,000

- If you are indifferent between A and B, you are **risk-neutral**
- If you prefer B, you are **risk-seeking**
- If you prefer A, you are **risk-averse**

Most people are risk averse, and (‘relative’) risk aversion declines with wealth

Humans are generally risk averse

- A human that is risk-averse would reject investment portfolios that are fair or worse.
- Risk-aversion penalizes the expected rate of return in a risky portfolio.

- Investors need to be paid a **Risk Premium** to compensate for owning risky assets.

More risk -> higher expected excess returns.

Statistics

Statistical Theory

- **Random Variable:** A variable that takes different values with some probability

Values	0	1	2
Probability	0.25	0.50	0.25

- **Probability Distribution:** Schedule of all possible payoffs and probabilities

Population Mean and Variance

X_i	0	1	2
Probability (p_i)	**0.25**	**0.50**	**0.25**

- **Expected Value (mean):** Expected payoff of a random variable

  $E(X) = \sum_{i=1}^{N} X_i p_i$ where X is outcome in state i, probability of state i

- **Variance:** Expected value of squared distance from the expected payoff. (variability)

  $$\sigma^2 = VAR(X) = \sum_{i=1}^{N} (X_i - E(X))^2 p_i$$

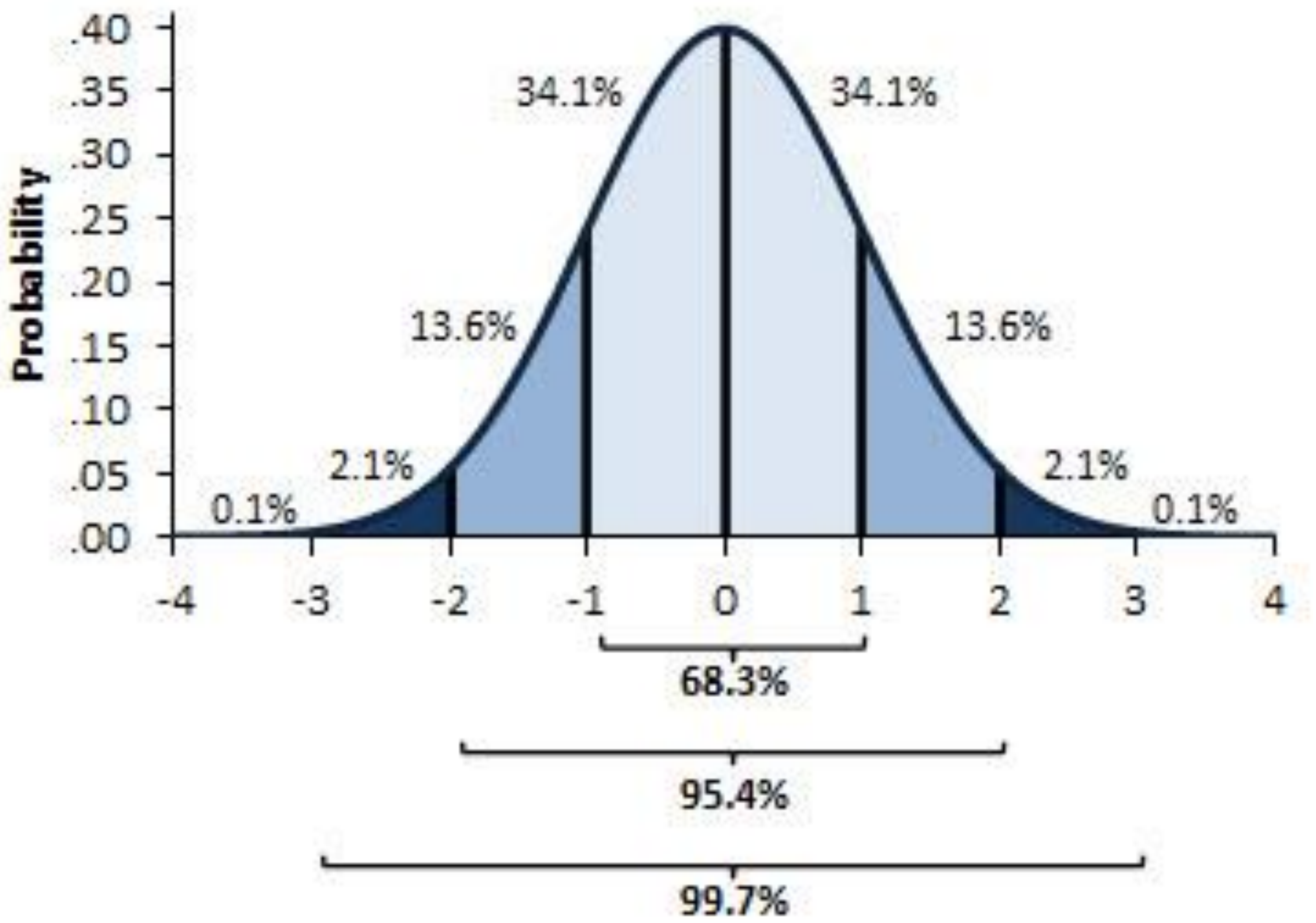
  where X is outcome in state i, probability of state i

- **Standard Deviation (s.d.):** $s = \sigma = \sqrt{\sigma^2}$

Standard Deviation

Standard Deviation (s.d.): $s = \sigma = \sqrt{\sigma^2}$

Standard deviation of a normal distribution



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Population Covariance

State	1	2	3
X	0	1	2
Y	2	1	0
Probability (p_i)	0.25	0.50	0.25

- **Covariance:** Co-movement between two random variables (X and Y)

$$COV(X,Y) = \sum_{i=1}^{N} (X_i - E(X))(Y_i - E(Y))p_i$$

$VAR(X,Y)$ =VAR(X)+VAR(Y)+ 2 COV (X,Y)

	X	Y
X	COV(X,X)=VAR(X)	COV(X,Y)=COV(Y,X)
Y	COV(Y,X)=COV(X,Y)	COV(Y,Y)=VAR(Y)

State	1	2	3
X	0	1	2
Y	2	1	0
Probability (p_i)	0.25	0.50	0.25

Correlation Coefficient

- **Correlation Coefficient:** Standardized measure of co-movement between two random variables (X and Y)

$$\rho_{XY} = \frac{COV(X,Y)}{\sigma_X \sigma_Y}$$

- If correlation coefficient = 0, then **uncorrelated**
- If correlation coefficient = **+1, then perfectly positively correlated**
- If correlation coefficient = **-1, then perfectly negatively correlated**

From Population (Theory) to Samples (Practice)

We don't know underlying probabilities of events (p_i) in sample data.

We instead give each observation a weight of 1/N, where data has N observations.

	Population	Samples
Expected Value	$E(X) = \sum_{i=1}^{N} X_i p_i$	$\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$
Variance	$\sigma^2 = \sum_{i=1}^{N} (X_i - E(X))^2 p_i$	$s^2 = \frac{1}{N-1}\sum_{i=1}^{N} (X_i - \bar{X})^2$
Covariance (p_i)	$COV(X,Y) = \sum_{i=1}^{N} (X_i - E(X))(Y_i - E(Y))p_i$	$COV(X,Y) = \frac{1}{N-1}\sum_{i=1}^{N} (X_i - \bar{X})(Y_i - \bar{Y}))$

Returns and Risk on Multiple Stocks

Returns on Multiple Stocks

- "Portfolios" are the combination of securities held (e.g. multiple stocks, stocks+bonds)
- Expected return of two securities is the weighted average of both returns:

$$E[R_{portfolio}] = w_A \times E[r_A] + w_B \times E[r_B]$$

Where $w_A + w_B = 1$ are the weights of two securities in the portfolio.

Calculate the returns on this rodeo stock portfolio

Expected return to Reba Stock is 10%

Expected return to Bun B Stock is 5%

Image: Courtesy Houston Livestock Show and Rodeo

a) What's the return to a portfolio investing 75% in Reba Stock and 25% in Bun B Stock?

b) What's the return to a portfolio investing 50% in Reba Stock and 50% in Bun B Stock?

c) What's the return to a portfolio investing 0% in Reba Stock and 100% in Bun B Stock?

Calculate the returns on this rodeo stock portfolio

Expected return to Reba Stock is 10%

Expected return to Bun B Stock is 5%

a) What's the return to a portfolio investing 75% in Reba Stock and 25% in Bun B Stock?

0.75 x 10% + 0.25 x 5% = 8.75%

b) What's the return to a portfolio investing 50% in Reba Stock and 50% in Bun B Stock?

0.50 x 10% + 0.50 x 5% = 7.50%

c) What's the return to a portfolio investing 0% in Reba Stock and 100% in Bun B Stock?

0 x 10% + 1 x 5% = 5.00%

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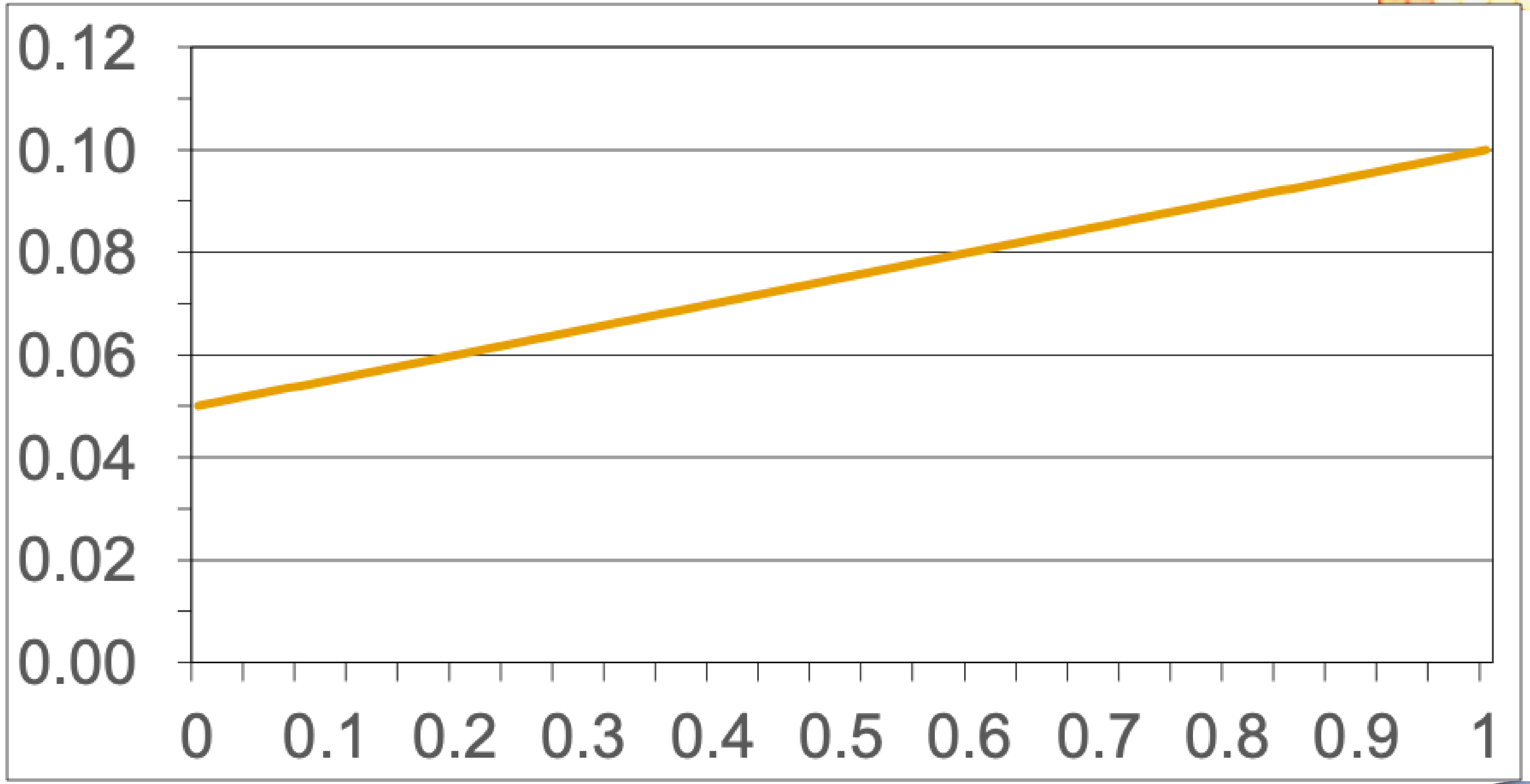
Returns on this rodeo stock portfolio

Expected returns to Reba Stock is 10%, & Bun B Stock is 5%

Portfolio Expected Returns

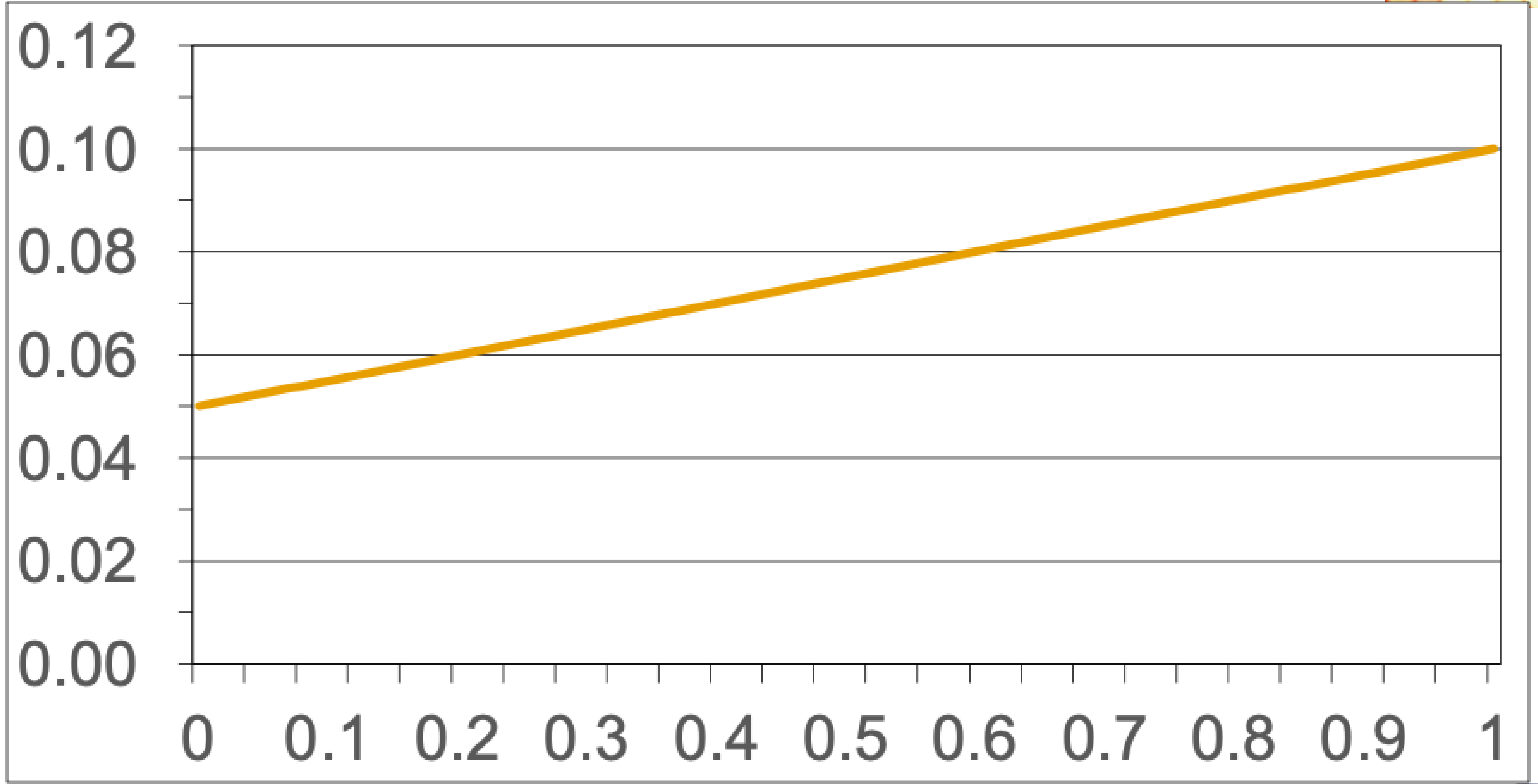


Fraction of Reba Stock in Portfolio

Clearly Reba Stock is higher returns, but putting 100% into this may be risky business

Expected returns to Reba Stock is 10%, & Bun B Stock is 5%

Portfolio Expected Returns



Fraction of Reba Stock in Portfolio

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Standard Deviation in Returns on Multiple Securities - Example with two securities A and B

Standard deviation (σ) is the measure of riskiness that we use most often for portfolios

$$\sigma^2_{Portfolio} = VAR(Portfolio) = VAR(w_A r_A + w_A r_A)$$

$$\sigma^2_{Portfolio} = VAR(w_A r_A) + VAR(w_B r_B) + 2\,COV(w_A r_A + w_B r_B)$$

$$\sigma^2_{Portfolio} = w_A^2 VAR(r_A) + w_B^2 VAR(r_B) + 2 w_A w_B COV(r_A + r_B)$$

$$\sigma^2_{Portfolio} = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B\, \rho_{A,B} \sigma_A \sigma_B$$

$$\sigma_{Portfolio} = \sqrt{w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B\, \rho_{A,B} \sigma_A \sigma_B}$$

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$$\sigma^2_{Portfolio} = w_A^2 VAR(r_A) + w_B^2 VAR(r_B) + 2 w_A w_B COV(r_A + r_B)$$

$$\sigma^2_{Portfolio} = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B\,\rho_{A,B}\sigma_A\sigma_B$$

$$\sigma_{Portfolio} = \sqrt{w_A^2\sigma_A^2 + w_B^2\sigma_B^2 \boxed{+ 2w_A w_B\,\rho_{A,B}\sigma_A\sigma_B}}$$

Covariance term is important!

The risk of a portfolio is ***NOT*** just the weighted average of standard deviations of each security in portfolio.

Standard Deviation in Returns on Multiple Securities - What if more than two securities?

2 Securities (A and B): $\sigma_{Portfolio\_2} = \sqrt{w_A^2 VAR(r_A) + w_B^2 VAR(r_B) + 2w_A w_B COV(r_A, r_B)}$

3 Securities (A, B, and C): $\sigma_{Portfolio\_3} =$

$$\sqrt{w_A^2 VAR(r_A) + w_B^2 VAR(r_B) + w_C^2 VAR(r_C) + 2w_A w_B COV(r_A, r_B) + 2w_A w_C COV(r_A, r_C) + 2w_B w_C COV(r_B, r_C)}$$

4 Securities (A, B, C and D): $\sigma_{Portfolio\_4} =$

$$\sqrt{\begin{gathered} w_A^2 VAR(r_A) + w_B^2 VAR(r_B) + w_C^2 VAR(r_C) + w_D^2 VAR(r_D) + 2w_A w_B COV(r_A, r_B) + 2w_A w_C COV(r_A, r_C) + \\ 2w_A w_D COV(r_A, r_D) + 2w_B w_C COV(r_B, r_C) + 2w_B w_D COV(r_B, r_D) \end{gathered}}$$

With N securities make sure you are including the variances for each security ***and*** also all the covariances.

Calculate the standard deviation in returns on this rodeo stock portfolio

Expected return to Reba Stock is 10% & standard deviation 20%.

Expected return to Bun B Stock is 5% & standard deviation 10%.

Correlation coefficient between Reba Stock and Bun B Stock is 0.5.

What's the standard deviation in returns to a portfolio investing 75% in Reba Stock and 25% in Bun B Stock?

Calculate the standard deviation in returns on this rodeo stock portfolio

Expected return to Reba Stock is 10% & standard deviation 20%.

Expected return to Bun B Stock is 5% & standard deviation 10%.

Correlation coefficient between Reba Stock and Bun B Stock is 0.5.

What's the standard deviation in returns to a portfolio investing 75% in Reba Stock and 25% in Bun B Stock?

$w_A = 0.75, \quad w_A = 1 - 0.75 = 0.25$

$\sigma_A = 0.2, \sigma_B = 0.1 \quad \rho_{A,B} = 0.5$

$$\sigma_{Portfolio} = \sqrt{w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B\, \rho_{A,B}\sigma_A\sigma_B}$$

$$\sigma_{Portfolio} = \sqrt{0.75^2 \times 0.2^2 + 0.25^2 \times 0.1^2 + 2 \times 0.75 \times 0.25 \times 0.5 \times 0.2 \times 0.1}$$

$$\sigma_{Portfolio} = \sqrt{0.026875} = 0.1639 \text{ or } 16.39\%$$

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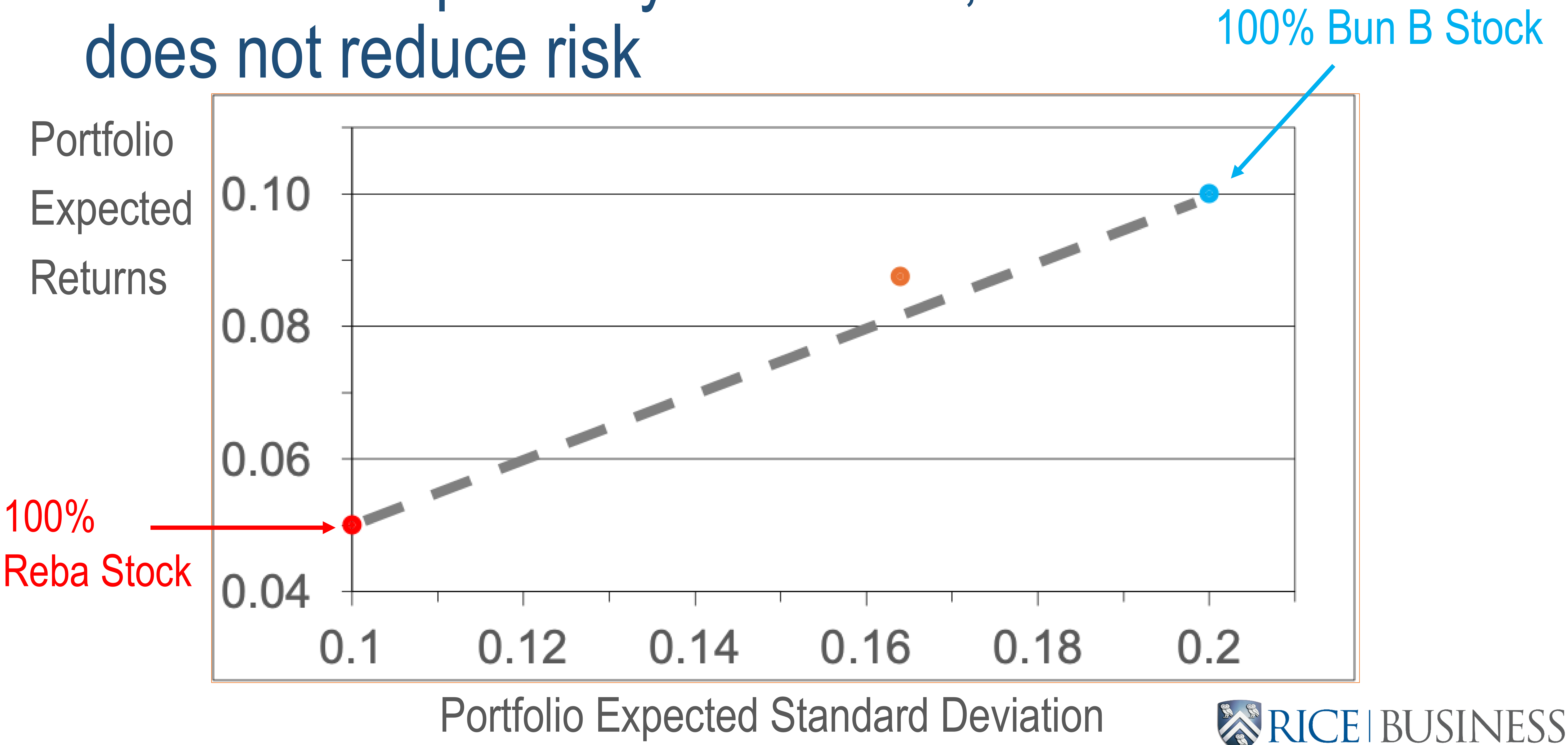
If stocks are perfectly correlated, diversification does not reduce risk



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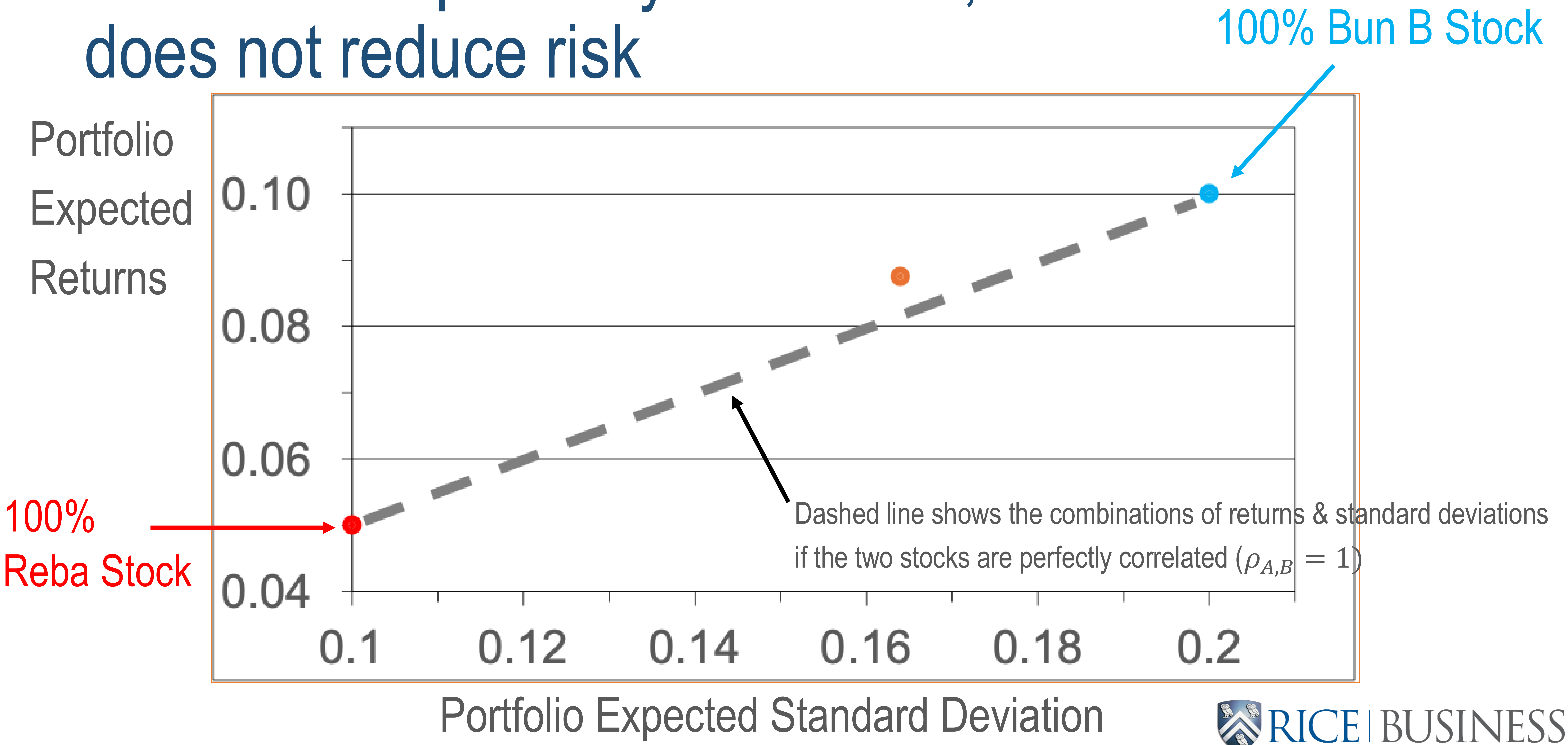

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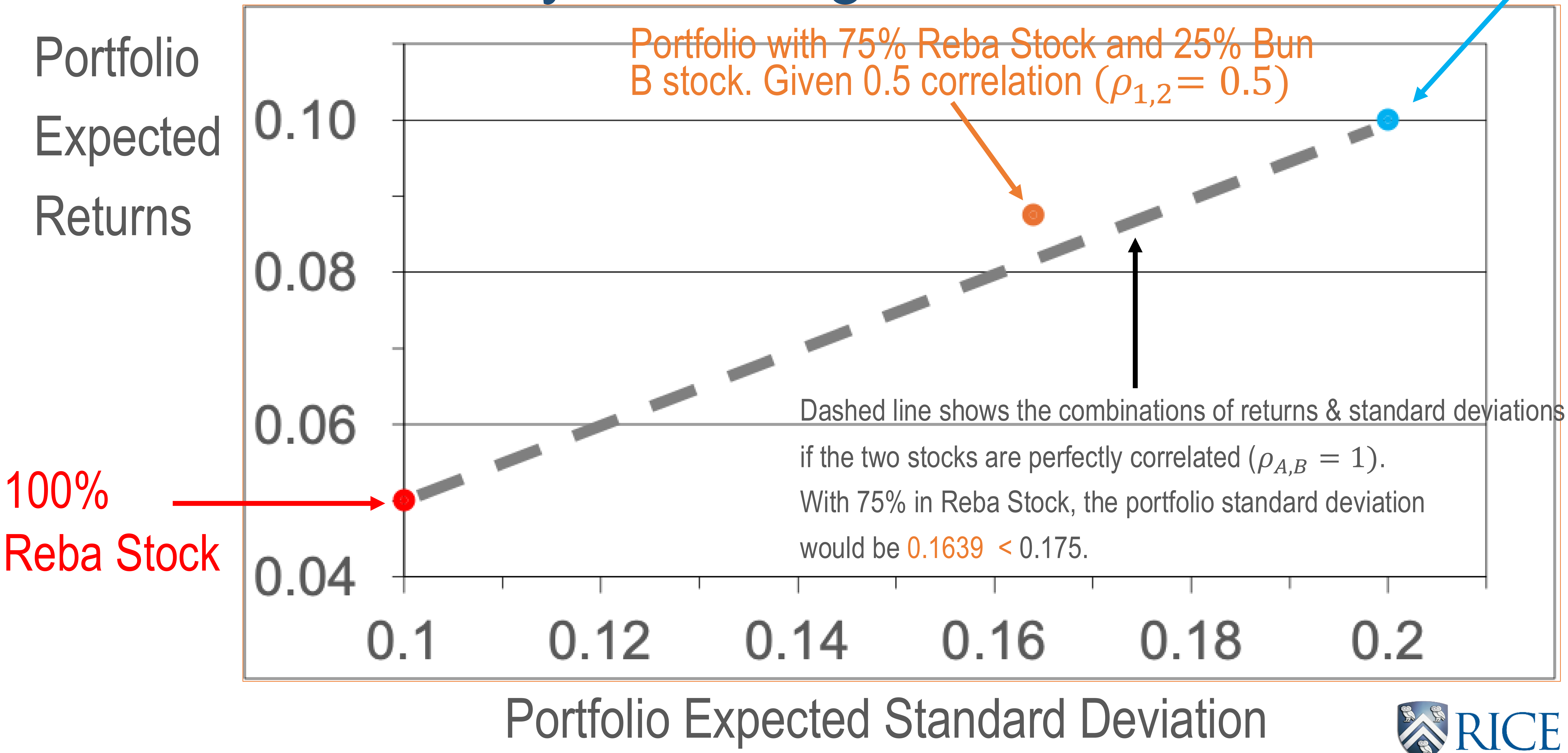
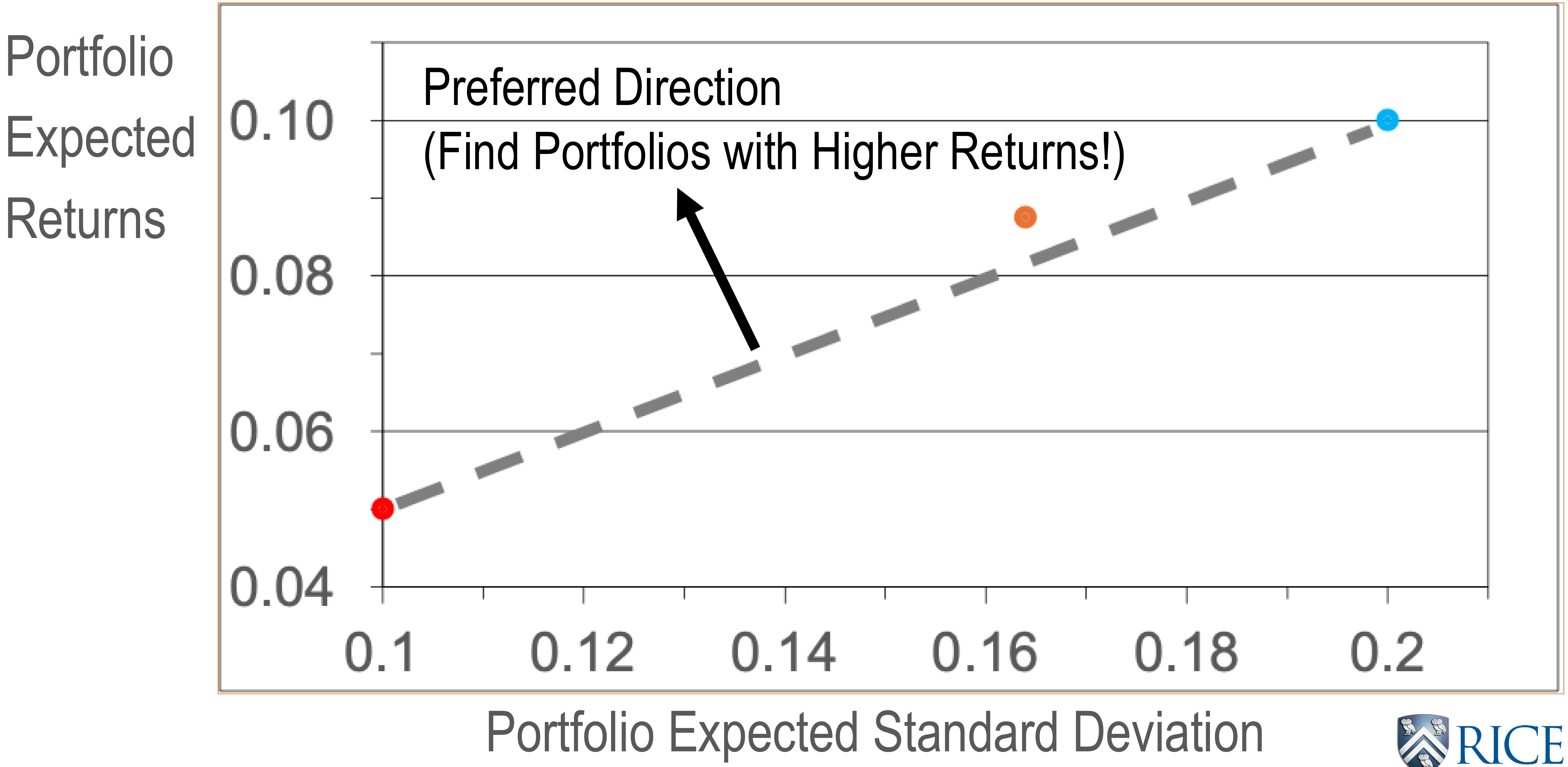
If stocks are ***not*** perfectly correlated, diversification has benefits by reducing risk

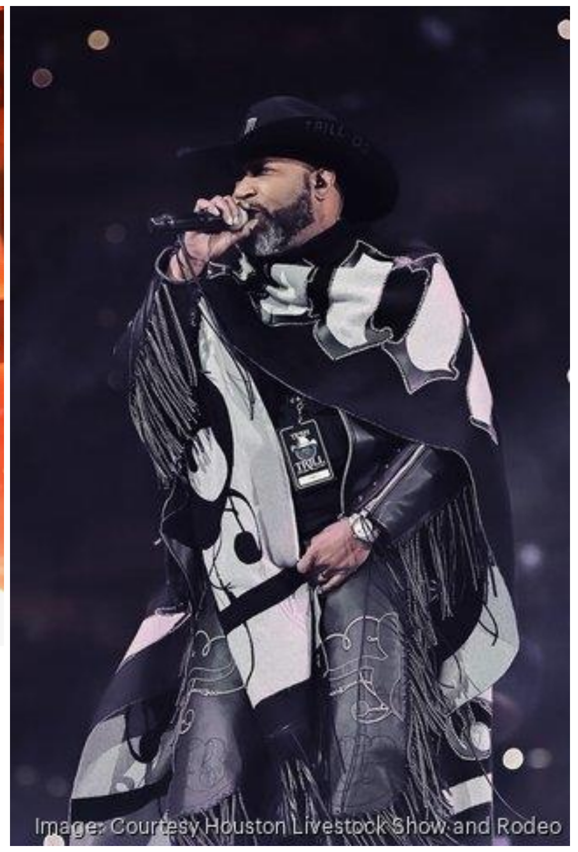


Portfolio Theory & Efficient Frontiers

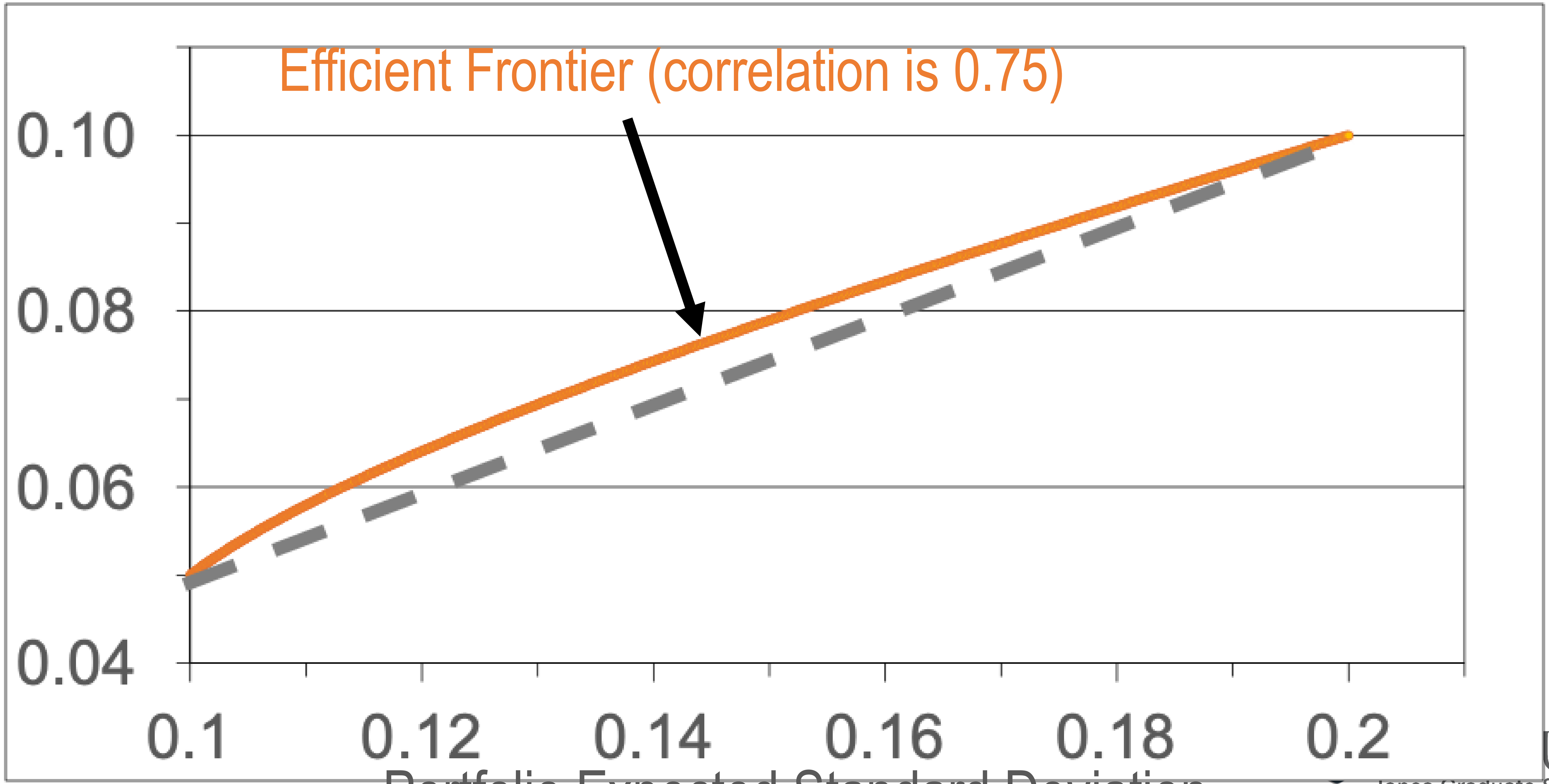
Benefits to Diversification



Efficient Frontier for our Reba and Bun B stock portfolio assuming correlation is 0.75

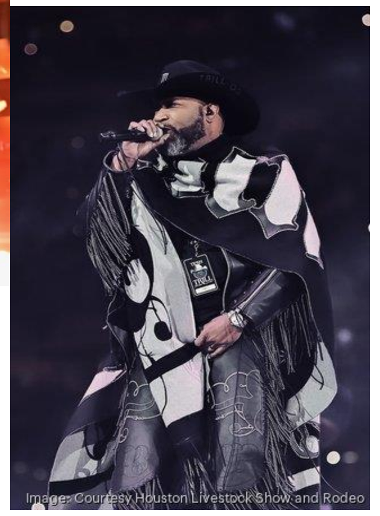


Portfolio Expected Returns



Portfolio Expected Standard Deviation

Returns and Risk for Reba and Bun B stock portfolio at different correlations



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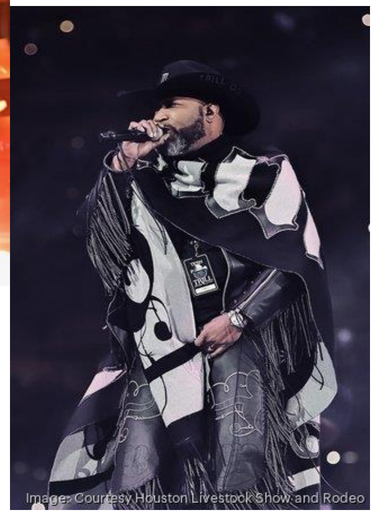
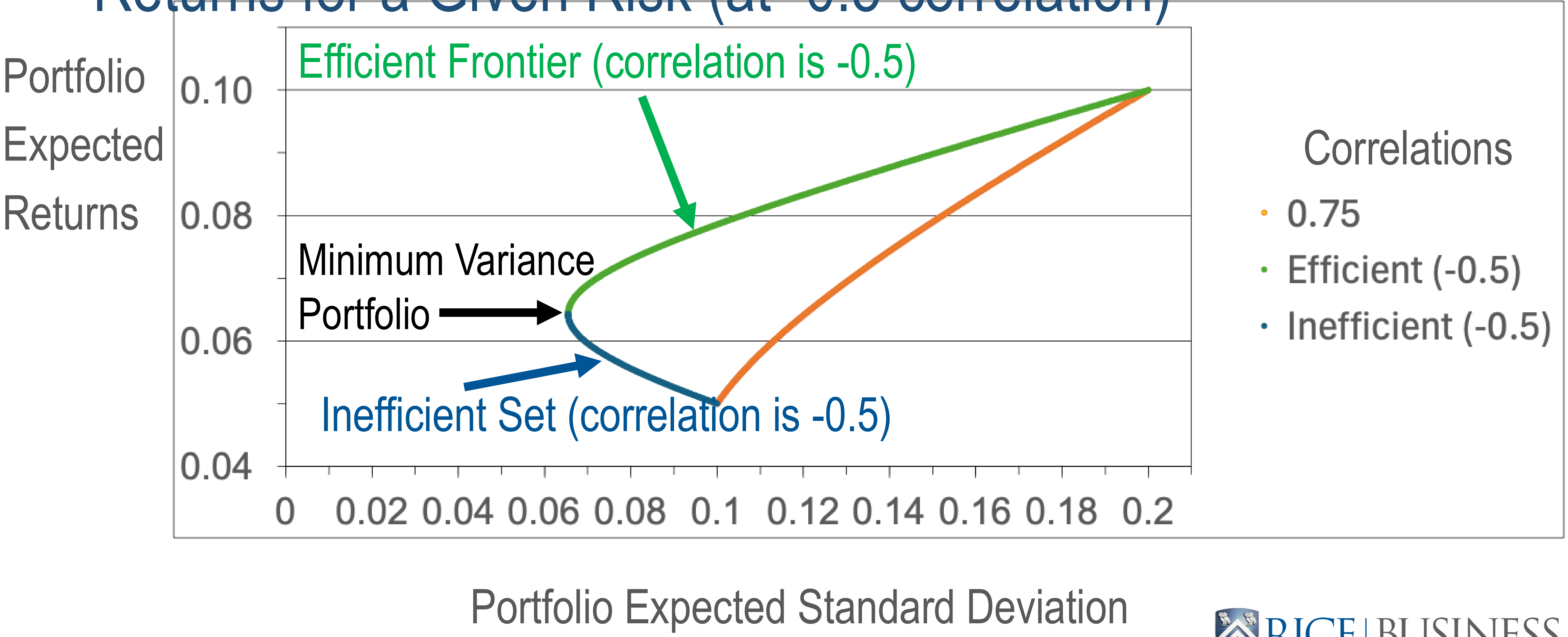


Finding Frontier of Portfolios With Highest Expected Returns for a Given Risk (at -0.5 correlation)



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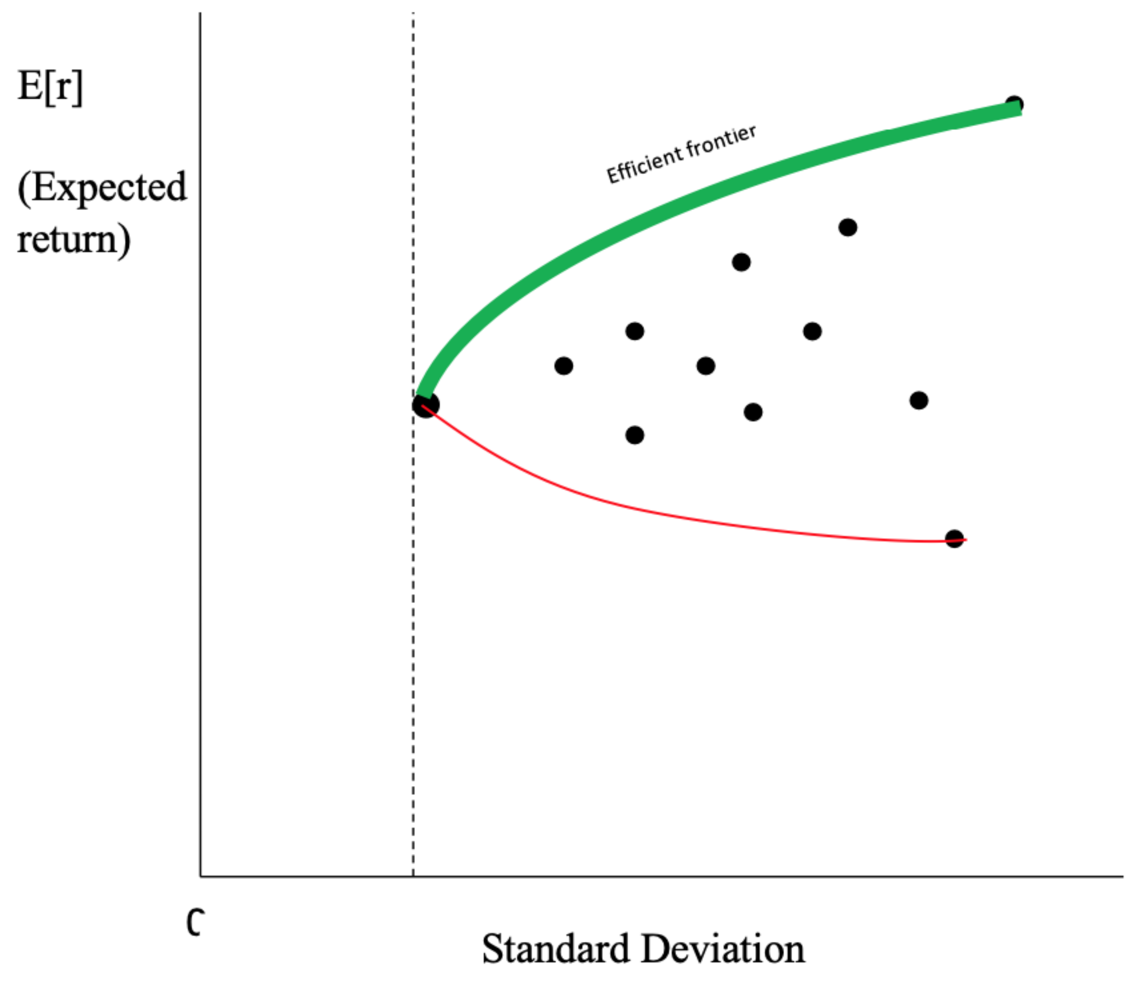
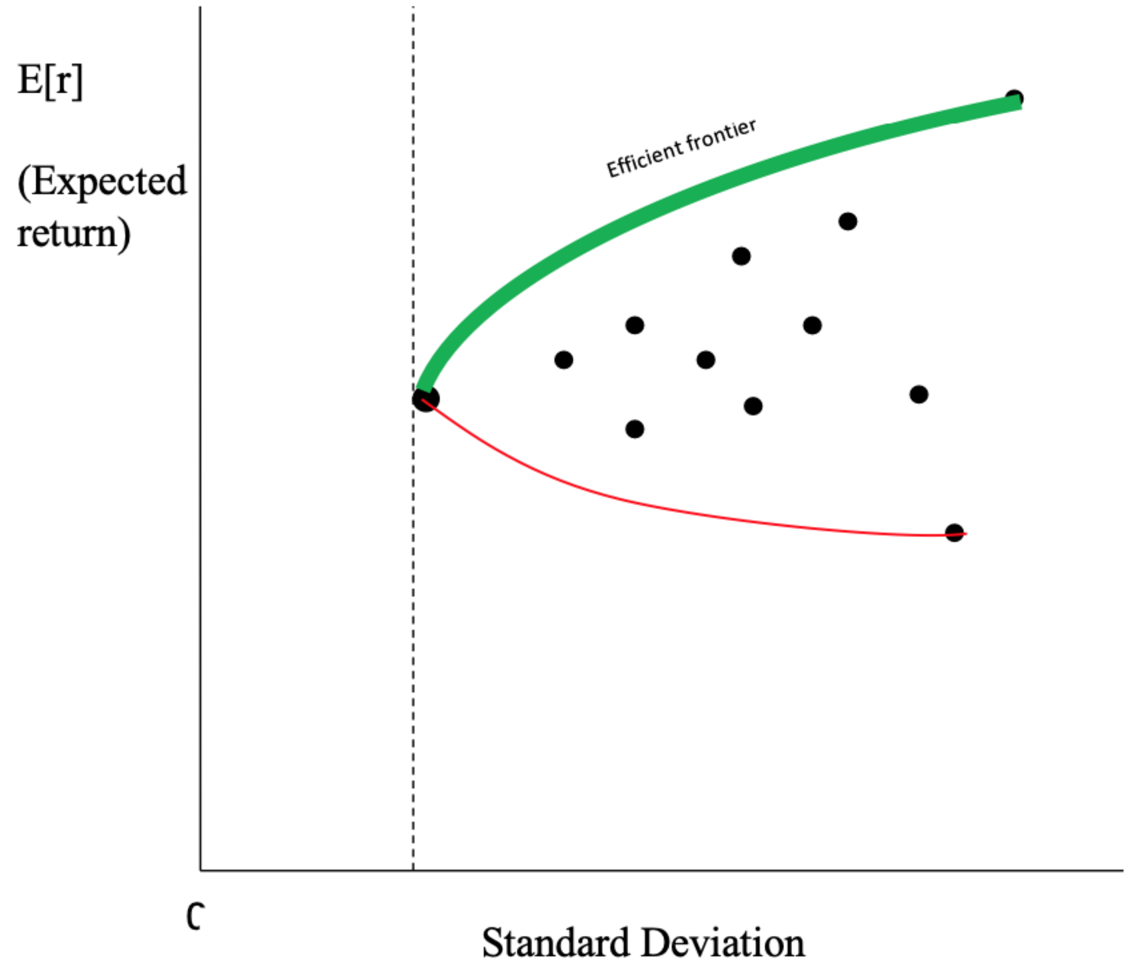
Efficient Portfolios



- A portfolio of securities can have a lower standard deviation than any of the individual securities in a portfolio.
- (Rational) Investors should choose a portfolio in the efficient set.

Efficient Portfolios



- A portfolio of securities can have a lower standard deviation than any of the individual securities in a portfolio.
- (Rational) Investors should choose a portfolio in the efficient set.
- Efficient portfolios have:
  - The largest expected return for a given risk
  - The smallest risk for a given expected return
- **"Portfolio Theory"** maximize expected return for given level of risk (variance)
- Which efficient portfolio should an investor choose?
  - It depends on their risk tolerance!
  - Higher expected return portfolios carry more risk (**"Risk-Return Trade-off"**)

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Limits of Diversification

Portfolios

- Portfolio returns combine linearly from the securities.
  - They are a weighted average of expected returns on each.
- Risk (standard deviations) do not generally combine linearly.
  - They depend on the correlations (covariances) between securities.
  - Adding more securities can improve risk/return trade-off.

- If holding many securities, then portfolio risk is more dependent on the covariances between individual securities than the variances of the individuals. For example, Imagine portfolio equally-weighted across N stocks. Then the portfolio variance is:

$$= N\left(\frac{1}{N}\right)^2 Average\,Variance + N(N-1)\left(\frac{1}{N}\right)^2 Average\,Covariance$$

How Many Stocks Needed in Portfolio to Diversify Risk?

- Imagine all stocks have an expected variance of 0.4
- Each pair of stocks have a correlation of 0.7
- So $COV(i,j) = 0.7 \times \sqrt{0.4} \times \sqrt{0.4} = 0.28$ for any two stocks i and j

Number of Stocks in Portfolio	Portfolio Variance	Portfolio Variance Calculation
1	0.40	
2	0.34	
3	0.32	
100	0.28	
1,000	0.28	
1,000,000	0.28	

Why May Returns on Two Securities be Correlated?

A stock's price is the present value of future free cash flows (dividends)

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Changes in cash flows or discount rates change stock prices.

Common changes in these can be a source of correlation:

- **Common cash flow shocks**
  - Changes in input prices (e.g., energy) affect many companies in same way
- **Common discount rate shocks**
  - Changes in interests rates affect many companies in same way

Not All Risks Can Be Eliminated By Adding More Stocks to Portfolios

Total Risk = Non-Diversifiable Risk + Diversifiable Risk

Total Risk = Systemic Risk + Idiosyncratic Risk

Total Risk = Market Risk + Firm-Specific Risk

Non-Diversifiable Risks	Diversifiable Risks

Investors should only expect higher returns from bearing higher non-diversifiable risks.

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Non-Diversifiable Risks	Diversifiable Risks
Economic expansions	Fire destroying plant
Recessions / Macro Shocks	CEO unexpectedly leaves
Large Scale Events (e.g., Wars)	Government/regulator unexpectedly does not approve product (e.g., FDA drug approval)

Investors should only expect higher returns from bearing higher non-diversifiable risks.

Examples

Q1. Given this information, calculate…

Outcome	Probability	Return on Stock A	Return on Stock B
Boom	0.25	20%	5%
Normal	0.50	10%	10%
Bust	0.25	0%	15%

(a) Expected Return A:

(b) Expected Return B:

(c) Standard Deviation A:

(d) Standard Deviation B:

(e) Covariance (A,B):

Q1. Given this information, calculate…

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(a) Expected Return A:

0.25 x 0.20 + 0.50 x 0.1 + 0.25 x 0 = 10%

(b) Expected Return B:

0.25 x 0.05 + 0.50 x 0.1 + 0.25 x 0.15 = 10%

(c) Standard Deviation A:

Var(A)=0.25 x (0.2 - 0.1)^2 + 0.5 x (0.1 - 0.1)^2 + 0.25 x (0 - 0.1)^2 =0.005. Take square root. S.D.(A)=7.07%

(d) Standard Deviation B:

Var(B) = 0.25 x (0.05 - 0.1)^2 + 0.5 x (0.1 - 0.1)^2 + 0.25 x (0.15 - 0.1)^2 =0.00125. Take square root. S.D. (B) = 3.54%

(e) Covariance (A,B):

Cov(A,B) = 0.25 x (0.2 - 0.1) x (0.05 - 0.1) + 0.5 x (0.1 - 0.1) x (0.1 - 0.1) + 0.25 x (0 - 0.1) x (0.15 - 0.1) = -0.0025.

Correlation: -0.0025 / (0.0707 x 0.0354) = -1

Q1. Given this information, calculate…

Outcome	Probability	Return on Stock A	Return on Stock B
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Normal	0.50	10%	10%
Bust	0.25	0%	15%

Portfolio Benedict:

$150 in Stock A, $150 in Stock B.

Portfolio GK:

$100 in Stock A, $200 in Stock B.

(f) Expected Return & Standard Deviation of Portfolio Benedict:

E(r) = (150/300) x 0.1 + (150/300) x 0.1 = 10%

VAR = (150/300)^2 x (0.005) + (150/300) x (0.00125) + 2 x (150/300) x (150/300) x (-0.0025) = 0.0003125

S.D. = 1.77%

(g) Expected Return & Standard Deviation of Portfolio GK:

E(r) = (100/300) x 0.1 + (200/300) x 0.1 = 10%

VAR = (100/300)^2 x (0.005) + (200/300) x (0.00125) + 2 x (100/300) x (200/300) x (-0.0025) = 0%

S.D. = 0%

(h) Which Portfolio do you prefer? **Portfolio GK! It delivers same return but lower standard deviation!**

Q2. Use excel to calculate the covariance and correlation coefficient between returns on large and small stocks.

Year	Large Stocks	Small Stocks
2010	30.24	21.63
2011	28.78	-1.86
2012	20.76	31.51
2013	-7.93	1.26
2014	-11.72	24.56
2015	-22.77	-13.79
2016	25.50	52.77
2017	10.27	17.45
2018	5.61	5.96
2019	14.51	18.64